

Héctor Laria

Applied Scientist · ML Engineer

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SUMMARY

Applied Scientist and ML Engineer with a PhD in Computer Vision and 6+ years of hands-on deep learning experience across industry-funded projects, large-scale model training, and peer-reviewed research (CVPR, ICCV, ECCV, WACV). Core expertise in generative AI, diffusion models, fine-tuning, and model adaptation, with published work on model robustness and on evaluating the unintended effects of adaptation. Independently delivered end-to-end ML solutions in industry collaborations with Audi and Huawei R&D. Experienced coordinating multi-contributor work under hard deadlines: task assignment, timeline estimation, and distilling technical results for stakeholders (agile-style delivery applied to research). Strong software engineering background: fluent in Python, PyTorch, and Hugging Face ecosystems. Native Spanish speaker, professional-level English (MSc, PhD, and postdoc conducted in English).

EXPERIENCE

Banco Santander

AI Scientist, Responsible AI – Data & AI Science (DAISci)

Madrid, Spain

Sep 2026 – Present

Computer Vision Center (CVC)

Postdoctoral Researcher

Barcelona, Spain

Aug 2025 – Feb 2026

- Mentored PhD students on two research lines (fine-tuning drift and open-world forgetting in diffusion models), co-authoring two papers, since accepted at ECCV 2026 and an ECCV 2026 workshop (oral).
- Advised and led HPC infrastructure migration (user account servers, shared filesystems); transitioned cluster administration to the incoming team.

Computer Vision Center (CVC)

PhD Researcher

Barcelona, Spain

Sep 2020 – Jul 2025

Audi – Image-to-Video *Industry Project*

- Sole ML researcher on an industry feasibility study for image animation using early diffusion models; independently scoped the problem, designed the architecture, and delivered a Dockerized end-to-end inference pipeline as the final deliverable.
- Subsequently supervised an intern through the project's continuation and completion.

Huawei R&D – Deep Lifelong Learning *Industry Project*

- Contributed to a 2+ year industry collaboration with Huawei R&D; presented and discussed transfer learning and continual learning research in monthly meetings with a cross-company team of ~10 researchers.
- Collaboration produced two peer-reviewed publications (CVPR 2022, ICCV 2021); industry funding from Huawei enabled the research group to acquire dedicated GPU compute infrastructure.

ColorWave *WACV 2026 · github.com/hecoding/ColorWave*

- Developed ColorWave, a training-free method for precise RGB-level color control in text-to-image diffusion models (PyTorch, Hugging Face Diffusers); works by rewiring semantic attribute bindings inside IP-Adapter cross-attention (zero inference overhead over the base model).
- Outperforms prior fine-tuning-based methods in color accuracy across diverse object categories; compatible with Stable Diffusion (SDXL, SD3), and FLUX.

Open-World Forgetting & DriftScope *ECCV 2026 · ECCV 2026 Workshop (oral)*

- Coined and first to systematically analyze open-world forgetting in diffusion model fine-tuning: demonstrated that adapting a model to as few as 5 images causes semantic accuracy drops of up to 60% on unrelated concepts.
- Proposed a functional regularization strategy that significantly reduces both semantic and appearance drift while preserving generation quality.
- Follow-up work (DriftScope, ECCV 2026) developed a prompt-engineering methodology to identify the specific tokens driving the most drift in fine-tuned models, enabling targeted mitigation.

Hyper-Modulation *CVPR Workshop 2022 · github.com/hecoding/Hyper-Modulation*

- Built a transfer learning method to adapt pretrained unconditional GANs to conditional generation under data-scarce settings, with a self-initialization procedure requiring no real data.

HPC Infrastructure & MLOps

- Administered the research group's HPC cluster: 8 compute nodes, 50 GPUs, 60 users. Provisioned and hardened nodes, diagnosed GPU hardware failures, managed distributed storage.
- Built monitoring and observability stack with Prometheus and Grafana; automated node provisioning with Ansible; managed job scheduling with Slurm.
- Led large-scale distributed training on national HPC clusters (MareNostrum, Spain; Leonardo, Italy) across hundreds of GPUs.

Aalto University

Helsinki, Finland

Project Employee – Machine Learning

Nov 2019 – Aug 2020

- Implemented a WaveNet-based generative model in PyTorch for room impulse response (RIR) synthesis conditioned on acoustic room properties, enabling audio simulation for arbitrary room configurations.
- Designed a multimodal joint embedding space for room properties and RIRs using a contrastive learning approach, conceptually equivalent to CLIP and developed ~6 months before its publication.

Aalto University

Helsinki, Finland

Research Assistant – Computer Vision & NLP

Apr 2018 – Aug 2019

- Led the team's entry to the TRECVID 2019 Video-to-Text challenge (NIST benchmark), applying reinforcement learning-based sequence optimization to improve captioning metrics (BLEU, METEOR, CIDEr, ROUGE), an early instance of RLHF-style training for NLP.
- Contributed to DeepCaption, an open-source image captioning framework: implemented RL training objective using PyTorch for more human-like caption generation.
- Participated in MeMAD (Horizon 2020 European project), an international research collaboration on multimodal audiovisual content accessibility.

Complutense University of Madrid

Madrid, Spain

Research Assistant – Artificial Intelligence

Oct 2016 – Jun 2017

- Analyzed behavioral traces from web-based games and built a centralized data collection pipeline used by the research group.

SKILLS

ML & Frameworks: Python, PyTorch, Hugging Face (Diffusers, Transformers), TensorFlow, JAX, scikit-learn, NumPy, Pandas

Generative AI & Deep Learning: Diffusion Models, GANs, Flow Matching, Transformers/ViTs, LLMs, Fine-tuning, Model Adaptation, Transfer Learning, Multimodal AI

NLP & Language: Large Language Models (LLMs), NLP, Prompt Engineering, Text-to-Image Generation, Image Captioning, Video Generation

MLOps & Infrastructure: Docker, AWS EC2, Git, CI/CD, REST APIs, MLflow, W&B, TensorBoard, Prometheus, Grafana, Ansible, HPC/Slurm, Linux (RHEL, Debian)

Responsible & Trustworthy AI: Model Robustness, Interpretability & Explainability, Benchmarking & Evaluation, Model Monitoring, Drift Detection, Catastrophic Forgetting, Controllability, Model Behavior Analysis

Research & Experimentation: Experimental Design, Model Evaluation, Benchmarking, Statistical Analysis, A/B Testing, Scalable AI Systems, End-to-end Delivery

Programming Languages: Python, C++ (Modern), Rust, C, JavaScript, Java

Human Languages: Spanish (native), English (professional)

EDUCATION

Universitat Autònoma de Barcelona

Barcelona, Spain

PhD in Computer Vision and Deep Learning

Sep 2020 – Jul 2025

Thesis: "Advancing Transfer Learning and Control of Generative Image Models"

Aalto University

Helsinki, Finland

MSc in Machine Learning, Data Science and Artificial Intelligence, Honours distinction

2019

Thesis: "Deep Reinforcement Sequence Learning for Visual Captioning"

SELECTED OPEN SOURCE

- **ColorWave**: Training-free RGB-level color control for diffusion models (WACV 2026) · github.com/hecoding/ColorWave
- **Hyper-Modulation**: Unconditional-to-conditional GAN transfer learning (CVPR Workshop 2022) · github.com/hecoding/Hyper-Modulation
- **PIQ (PyTorch Image Quality)**: Contributed 2 state-of-the-art perceptual quality metrics to the open-source library

SELECTED PUBLICATIONS

- **ECCV 2026**: DriftScope: Measuring the Hidden Effects of Diffusion Model Adaptation
- **ECCV 2026 Workshop (oral)**: Assessing Open-World Forgetting in Generative Image Model Customization
- **WACV 2026**: Leveraging Semantic Attribute Binding for Free-Lunch Color Control in Diffusion Models
- **CVPR 2022 Workshop**: Transferring Unconditional to Conditional GANs with Hyper-Modulation
- **ICCV 2021**: TransferI2I: Transfer Learning for Image-to-Image Translation from Small Datasets

Full list: scholar.google.com/citations?user=oifXN6kAAAAJ